Classifying galaxy morphology with a CNN

Galaxy10 DECaLS: baselines, a custom CNN, transfer learning, Grad-CAM

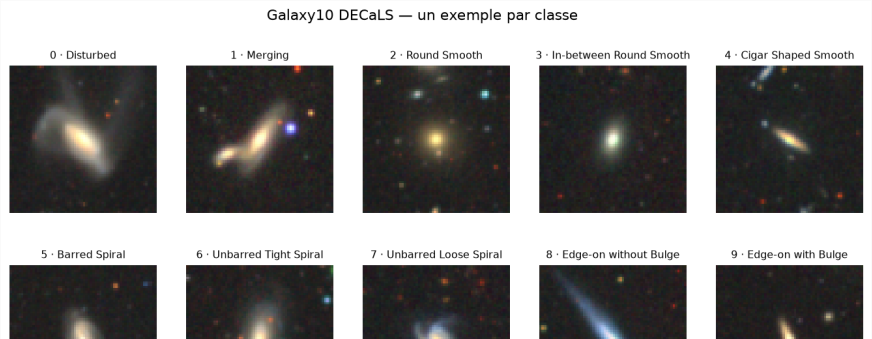
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The problem

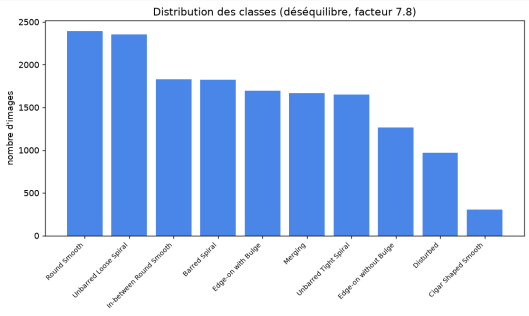
Galaxy Zoo asked **hundreds of thousands of volunteers** to sort galaxies by shape.

- 17,736 color images, 256×256, from the DESI Legacy Imaging Surveys.
- 10 morphological classes: smooth, spiral (barred / tight / loose), edge-on, merging.
- **Can a CNN do that job automatically?**



The data, and why it is hard

- Strong class imbalance (factor $\sim$8): Cigar Shaped 334 vs Round Smooth 2,645.
- Neighboring classes look alike: tight vs loose spirals, round vs cigar.
- Noisy labels (volunteer votes).
- Official split: test set kept untouched; validation is 15% of train.



Convolution in one slide

A convolution slides the **same small filter** over the whole image:

$$(I * K)_{i,j} = \sum_{m,n} I_{i+m,\,j+n}\, K_{m,n}.$$

- **Weight sharing**: one filter everywhere $\Rightarrow$ few parameters, a motif detector.
- **Translation equivariance**: move the pattern, the response moves with it.
- This is a **symmetry built into the architecture**, exactly the spirit of symmetries in physics. A galaxy has no preferred orientation, so we also augment with rotations and flips.

Our custom CNN

Four blocks `Conv 3x3` $\rightarrow$ `BN` $\rightarrow$ `ReLU` $\rightarrow$ `MaxPool`, channels 32/64/128/256, then adaptive pooling and a dense head.

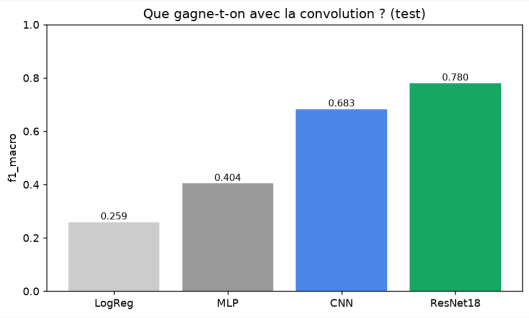
- Input $3 \times 64 \times 64$, halved at each pool.
- **1.44 M parameters**, 73% in the first dense layer.
- Weighted cross-entropy, Adam, cosine schedule, early stopping on macro-F1.

layer	output (C, H, W)
input	$3 \times 64 \times 64$
block 1	$32 \times 32 \times 32$
block 2	$64 \times 16 \times 16$
block 3	$128 \times 8 \times 8$
block 4	$256 \times 4 \times 4$
head	$\rightarrow 10$

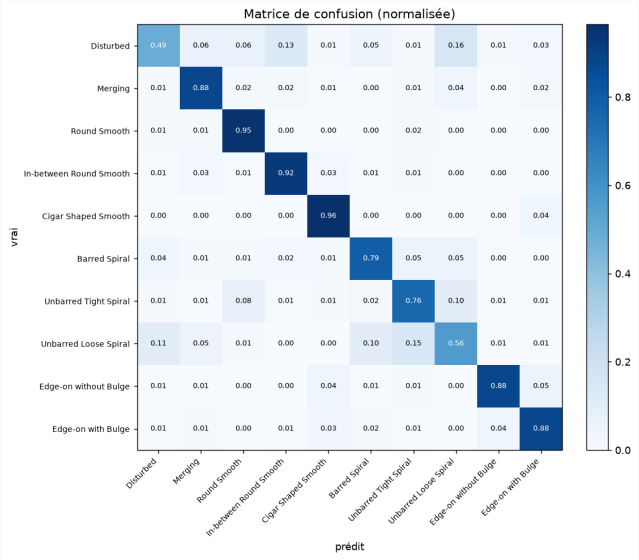
What does convolution buy us?

model	F1 macro
Logistic regression	0.26
MLP (pixels)	0.40
Custom CNN	0.68
ResNet18 (fine-tuned)	0.78

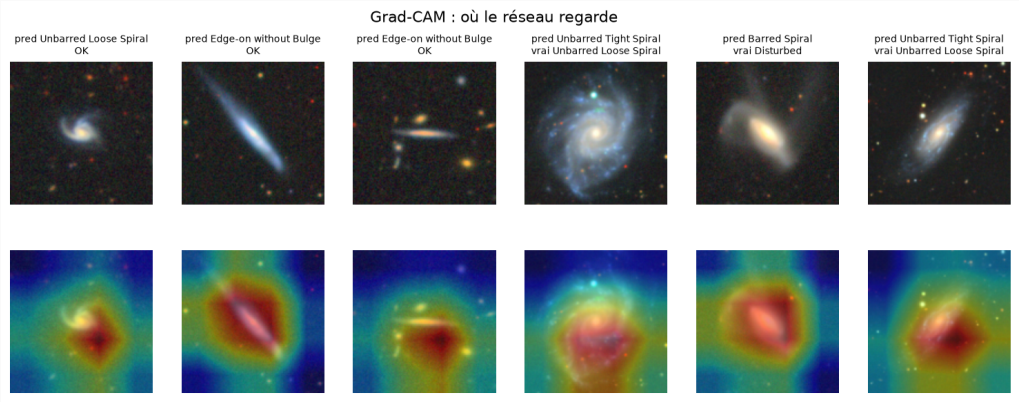
Baselines are blind to spatial structure. Convolution, then pretraining, explain the gain.



Where it fails: confusion matrix



Grad-CAM: where the network looks



On correct calls the heatmap sits on spiral arms, the bar, or the merging companion.
On errors it sometimes latches onto a neighbor or an artifact.

Ablations and limits

Ablations (macro-F1, test):

- full CNN: 0.68
- no augmentation: 0.69
- no BatchNorm: 0.72

Counterintuitive here: augmentation barely helps and removing BN even nudges F1 up. A low-overfit, single-seed regime. BN still speeds convergence.

Limits:

- noisy Galaxy Zoo labels cap accuracy;
- class imbalance;
- training cost (64/128 px, partial tuning).

Conclusion

- A CNN reproduces Galaxy Zoo's morphological sorting: ResNet18 reaches accuracy 0.80, macro-F1 0.78.
- The scale baseline $\rightarrow$ CNN $\rightarrow$ ResNet isolates the value of convolution.
- Grad-CAM shows the network uses real morphological structures.
- Key idea: convolution works on images because it **builds in a symmetry**, translation equivariance, the way physics builds in symmetries.

Thank you.